Waterworks

– Domestic water filtration empowered by acoustics.



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Abstract

Approximately 1,000 children under 5 years old die every day due to contaminated drinking water and a lack of basic sanitary facilities. Additionally, 1 million people die every year from lacking access to clean water. According to research, we in Sweden might have to rethink that this is something that only happens somewhere else. The climate crisis presents significant challenges to water filtration. Extreme weather events can contaminate water sources, and harmful algal blooms, saltwater intrusion, and toxins further complicate filtration. We will have to rethink how we take clean tap water for granted. This home water filtration unit, which uses acoustics to filter the water, examines our relationship to drinking water in Sweden and the habits around it, as well as the new behaviors and rituals that would emerge from this new normal.

Keywords

Human centered design, Speculative design, Design ecologies, Critical design, Industrial design, Product design, Water filtration, Household product, Tap water, Acoustic resonance, NASA.

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Introduction

What if we can't take our precious tap water for granted anymore? Maybe the centralized tap water systems work from time to time, and we need a backup to build resilience. Could a household filtration unit raise awareness regarding water scarcity and lacking access to clean drinking water?

Goal
The goal in this project is to improve the resilience in drinking water supply for the private user. I want to focus on an urban household. The target group will probably be a small family with one child living in Stockholm city centre. Water filtering solutions on the market will be investigated. The result will be some kind of water filtering unit implemented on the kitchen countertop in a household.

Purpose
An often-overlooked aspect of clean tap water is that we here in Sweden have an abundance of it and that there won't be any disturbance with the water distribution. The purpose with the project is to highlight the importance of clean and accessible tap water, and that we should start planning and prepare for a scenario where we don't have the same access to clean water as we do today.

Limitations
The intention is to limit the work to a private user living in Sweden. Lack of clean drinking water are common in many parts of the world, and this will be used as a foundation to reflect on our possible future in Sweden. Focus lies on Sweden since the topic usually gets neglected as something that happens somewhere else, and people generally don't think we are facing a water crisis.[1] The issue with water scarcity or contaminated water have many touchpoints and areas to improve. In this project the focus has deliberately been on a household product, since it has all the ingredients to address for a designer. The scale is right, and it incorporates a lot of challenges connected to choice of materials, what behaviours can be changed, who does users interact with the object etcetera.

Method
This project has been made by using a variety of methods, such as – literature studies, workshops, interviews, material studies, physical prototyping and 3D Modelling.

UN global goals
In 2015, leaders all over the world agreed to 17 global goals, officially referenced as the sustainable development goals. This thesis has focused on goal 6: Ensure access to water and sanitation for all.[2]

1 Havs- och Vattenmyndigheten, 2023-08-24, " Vattenbrist och torka – så påverkar det vattenmiljön", https://www.havochvatten.se/miljopaverkan-och-atgarder/miljopaverkan/vattenbrist/vattenbrist-och-torka---sa-paverkar-det-vattenmiljon.html#:~:text=Vattenbrist%20och%20torka%20kan%20orsaka,mängd%20organismer%20som%20förbrukar%20syre. (2024-06-08).
2 UN, https://www.un.org/sustainabledevelopment/water-and-sanitation/ (2024-06-07).

Design problem

One in three people globally do not have access to safe drinking water.[3] According to WHO 1 million people are estimated to die each year from diarrhea because of unsafe drinking-water, sanitation and hand hygiene. Approximately 1000 children under 5 year are dying every day from lacking clean drinking water and sanitation.[4] Research from Svenskt Vatten indicates that Sweden might experience disturbances in access to clean tap water due to the climate change.[5] Extreme weather events can contaminate sources. Harmful algal blooms, saltwater intrusion and toxins can further complicate filtration. Most of Swedish waterworks were built over 50 years ago and are in urgent need for massive upgrade and renovation.[6] Also, a growth in population constitutes a higher pressure on water and sewage services. With these facts in mind, we need to prepare for a future scenario where the water filtration of what we know of today is not satisfactory enough. If there is a major power outage the municipal water service won't work.[7]

Target group

The focus in this project will be on the Swedish market. Although it is important to know that this target group is not the group that is most affected by unclean drinking water. The decision to work with the Swedish user is because there is an interest for new innovations and high acceptance towards new technology among this group. [8]

Water filtration

The market
The private domestic water filtering market today is rapidly growing, driven by increasing awareness of water quality issues and health concerns. Brands like Brita and PUR dominate the market with affordable carbon filter pitchers, which remain popular for their ease of use. Advanced filtration systems such as the APEC Water Systems' reverse osmosis units and Aquasana's UV purification systems are becoming more prevalent among consumers seeking higher levels of filtration. Smart water filters, like those from the brand Hydros, offer real-time monitoring and connectivity features, enhancing safety. Additionally, companies like Aarke are gaining traction for their commitment to design and user-friendly filtration solutions, appealing to lifestyle conscious consumers. Despite their benefits, domestic water filtration systems have several downsides:

Advanced systems like reverse osmosis units from APEC Water Systems can be expensive to purchase and maintain, requiring regular filter replacements and potential professional installation. Systems such as those from Aquasana often require frequent maintenance, including replacing filters and cleaning components, which can be time-consuming and costly. While brands like Aarke focus on sustainability, many traditional systems, including Brita and PUR, generate plastic waste from disposable filters, contributing to environmen-

3 WHO, 2019, https://www.who.int/news/item/18-06-2019-1-in-3-people-globally-do-not-have-access-to-safe-drinking-water-unicef-who (2024-06-07).
4 WHO, https://www.who.int/news-room/fact-sheets/detail/drinking-water (2024-06-08).
5 Svenskt vatten, 2014, https://www.svensktvatten.se/globalassets/dricksvatten/ravatten/m135.pdf (2024-06-08).
6 Svensk vatten, 2020, page 18, https://www.svensktvatten.se/globalassets/om-oss/omvarldsanalys-verksamhetsstrategi-2020.pdf (2024-06-09).
7 MSB, https://www.msb.se/sv/rad-till-privatpersoner/forbered-dig-for-kris/ (2024-06-01).
8 Vinnova, 2023, https://www.vinnova.se/nyheter/2023/06/transformativ-teknik--en-grund-for-sveriges-framtida-konkurrenskraft/ (2024-05-29).

tal pollution. Basic carbon filter pitchers from brands like Brita and PUR may not effectively remove all contaminants, such as heavy metals, pharmaceuticals, and certain bacteria, limiting their overall effectiveness in ensuring water safety.



Picture 1: workshop at Konstfack.



Picture 2: lowtech water filter (outcome from the workshop).

Design process

Workshop 1
To get familiar with existing low tech filtering techniques an initial workshop was held at Konstfack with four participants. The task for them was to design a water filtering unit based on information given in the beginning of the session. The goal was to create a filtering unit that would work in an emergency.

Interview 1
After the initial workshop an interview were conducted 2024-02-14 with Carl Ljungh from Aarke at their office in Stockholm. Aarke is a Swedish company that produces water filtration jugs, carbonators, coffee brewers among other household products. Carl Ljungh has an extensive background in industrial design and product development. During the interview existing products and Aarkes in specific were discussed. According to Carl a lot of customers don't want to store their water in plastic containers or even have plastic that is in contact with the filtered water (due to risk of pollutants and microplastics).

The interview gave a lot of insights into the different criteria for a household product, such as the user journey, dimensions of the product, which materials to use, how to clean the filtration and the esthetics of the product.

Interview 2
A zoom-meeting was held 2024-02-17 with Clara Hermansson, head of product development at Alnarp Cleanwater. A company that offers solutions for filtering sewage water with natural processes using plants and living organisms, and several products for storing and collecting rainwater.

The interview highlighted the difficulties dealing and suppling drinking water in Sweden. Both from a logistical point of view, but also from a regulatory aspect. By law the municipalities can't supply clean drinking water to its citizens unless the water has been processed by official water treatment plants. Discussions were also held around emergency situations and how crisis planning affects the distribution of drinking water.



Picture 3. Workshop 2: New household water filtering products was ideated.

Workshop 2
With the learnings from the interviews and first workshop, a new workshop was held where 8 people were to ideate household water filtration products. During this workshop water was served from a natural well in Nackareservatet (Sandakällan). 3 people (out of 8) declined to taste the water, due to the fear of getting ill. One participant specifically mentioned her train travel later in the afternoon as an excuse not to taste the water. None got sick, but it highlighted the safety and trust aspect regarding drinking water. When camping, or in extreme situations one can accept higher risk-taking. But in a normal everyday situation the threshold towards catching an upset stomach is higher. A very valuable learning outcome. Other interesting discussion was the difficulty that would arise with a power outage, with the consequence that the municipal water would stop working. The participants in the workshop stressed the modular aspects and flexibility in a household water filtering product. Many existing products on the market today rely on functioning tap water which they then filter underneath the sink. Other filtering jugs are focusing on removing unwanted taste from the water (for example chlorine). But the downside with these is that they can't handle heavily contaminated water.

Problem formulation

The interviews and workshops had brought up several challenges with a domestic water filtration unit. The most crucial was safety and trust. How do you build trust with the user to rely on the filtration process? In Sweden hardly anyone questions the quality of the tap water. Is it possible to gain the same level of trust in a solitary filtering unit that the user has in his/her kitchen? The other problem that was identified from the workshops where the need for modularity and flexibility in the product. Even in a worst-case scenario as a full-scale war would be, there is an everyday life going on besides all the horrible. That means people will have to go to work, cook and make sure the kids go to school. Power will go on and off from day to day. This argues for a modular filtering product that functions with or without electricity and is trustworthy in its functionality.

Concepts and ideation



Picture 4. Concept 1 - Prototype of an acoustic resonance filtration unit. Approximately 50 cm in diameter and 60 cm high.

Concept 1 – Acoustic resonance filtration

According to MSB it is necessary to have minimum 2-3 litres per person/day.[9] In a household consisting of 2 adults and one child, the need would be somewhere between 6 and 9 litres per day as a minimum. This amount informed the size of the product. An initial prototype was produced to do further tests with users. Literature studies done simultaneously resulted in the finding of a new filtering technique. A filtration innovation developed for use in zero gravity (by NASA). An acoustically driven molecular sieve embedded with small-diameter carbon nanotubes. First, water enters the device and contacts the filter matrix. Carbon nanotubes within the matrix allow only water molecules to pass through, leaving behind any larger molecules and contaminants. The unique aspect of the technology is its use of acoustics to help drive water through the filter. The combination of acoustics and small-diameter carbon nanotubes in this innovation make it an effective and efficient means of producing contaminant-free, clean water.[10] This technique was hypothetically applied in the prototype

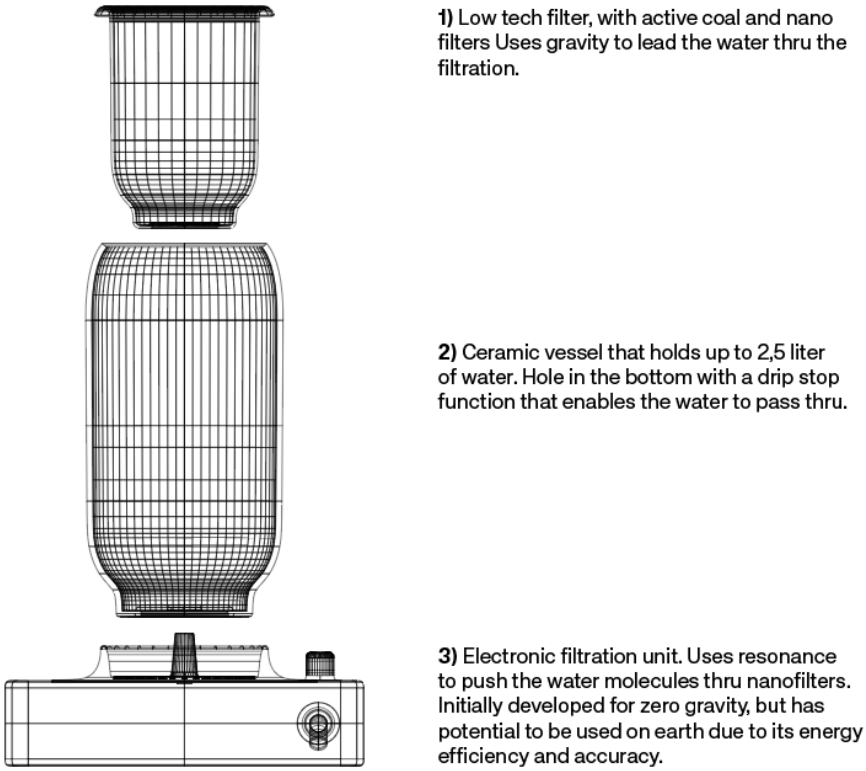
9 MSB, 2020, https://www.msb.se/sv/rad-till-privatpersoner/forbered-dig-for-kris/hemberedskap---preppa-for-en-vecka/vatten/ (2024-06-08).

10 NASA, ”Filtering Molecules with Nanotube Technology”, https://technology.nasa.gov/patent/MSC-TOPS-29 (2024-01-14).

(a Bluetooth speaker was mounted inside the prototype shown in picture 5) and tested in a workshop and discussed in an interview with Carl Ljungh at Aarke.

Evaluation

The feedback from the workshop was very positive and a lot of interest was focused on the new acoustic resonance technique. According to participants in the workshop the technique offered a new and existing way to filter water that they found intriguing. In the interview on the other hand, conducted 2024-04-01 with Carl at Aarke, another viewpoint was discussed. Carl mentioned that the prototype was too big to fit on a kitchen countertop (maximum hight from countertop to kitchen cabinet is usually 50 cm), and that the technology was maybe too complicated and unnecessary. The arguments circled around if the product could be made totally low-tech to meet the criteria of functioning under a power outage. The sustainability aspect was also discussed with the argument that a low-tech product could be more environmentally friendly if made from carefully sourced materials.



Picture 5. Schematic display of all the components in the design proposal.

Chosen direction – Concept 2

The above-mentioned feedback from workshop and interview resulted in a new design proposal. This time the size of the vessel that holds the water was downsized to be able to hold 2-3 litres of water (instead of 6-9 litres). This shrunk the dimensions significantly to a vessel that was 27 cm high and 15 cm in diameter. The vessel had a mechanical filter (nano filter and active coal filtration) to act as the first barrier of filtration. In a scenario of power outage this manual filtration relied on gravity to pull down the water thru different stages of filtrations. The result might be good enough in a crisis. When the user has electricity the water can travel to the next stage. That stage consisted of an acoustic resonance box that was designed as a simple square box/base for the vessel to fit in. The vessel had a hole in the

bottom with a drip-stop, and when the vessel was placed on top of the box a brace/stick pushed open the drip-stop mechanism and let the water flow downwards into the resonance filtration box. From there the water could be fetched similarly to a bag-in-box wine that you place on the edge of the kitchen countertop or a table. A simple LED-light indicated when the water was clean to be fetched.

Design proposal



Picture 6. Final design in salt glazed stoneware and recycled steel.

Final design
The final design was made from salt glazed ceramics and recycled metal. The total hight of the product was 38 cm and base 25 cm wide. The first ceramic vessel filters the water from the worst pollutants, and the second box uses acoustic resonance to process the water to contaminant-free, clean water. The system is according to NASA very energy efficient and self-cleaning.[11] The vessel holds in total 5 litres of water, but only 2,5 litres when in use. The water filters via gravity and nano filters down towards the acoustic box. The whole filtering process takes about 20 minutes from start till finish. The vessel can be topped up if there is a need to filter more water (if the first batch of water have been dropped away to give room for new refill).

Name
Waterworks - Domestic water filtration empowered by acoustics.
Waterworks typically refer to a system or facility for supplying, storing, and treating water.

11 NASA, ”Filtering Molecules with Nanotube Technology”, https://technology.nasa.gov/patent/MSC-TOPS-29 (2024-06-03).

It encompasses the infrastructure and machinery used to collect, purify, and distribute water to homes, businesses, and other users. The term can also mean starting to cry, especially to gain sympathy or attention. Additionally, it can refer to the organs of the body through which urine is passed.[12]

Material

During the interviews with Ljungh the material aspect was discussed. The initial prototype was made to mimic terracotta clay. The salt glazed technique was brought up as a as a subtle nod to the past when Sweden produced salt glazed water pipes, sinks, and the world-famous Höganäs mug. The salt glazed technique is known to be the toughest industrial glazing there is.[13] This led to a field trip to Wallåkra salt glazing facility in Skåne, were Åsa Orrmell had the kindness to let a few stone ware samples be burnt in her kiln. The process is that the kiln gets fired up 3 times per year, and then the kiln gets heated up to +1280 C degrees. At that point several dampers are opend and sea salt was thrown in. Due to the heat, the salt evaporates in a chemical reaction and creates this durable and appealing surface.



Picture 7. Final design, 3 units.

12 Oxford Learners Dictionaries, https://www.oxfordlearnersdictionaries.com/definition/english/waterworks#:~:text=%2Fw%C9%94%CB%90t%C9%99rw%C9%9C%CB%90rks%2F,supplying%20water%20to%20an%20area (2024-06-02).

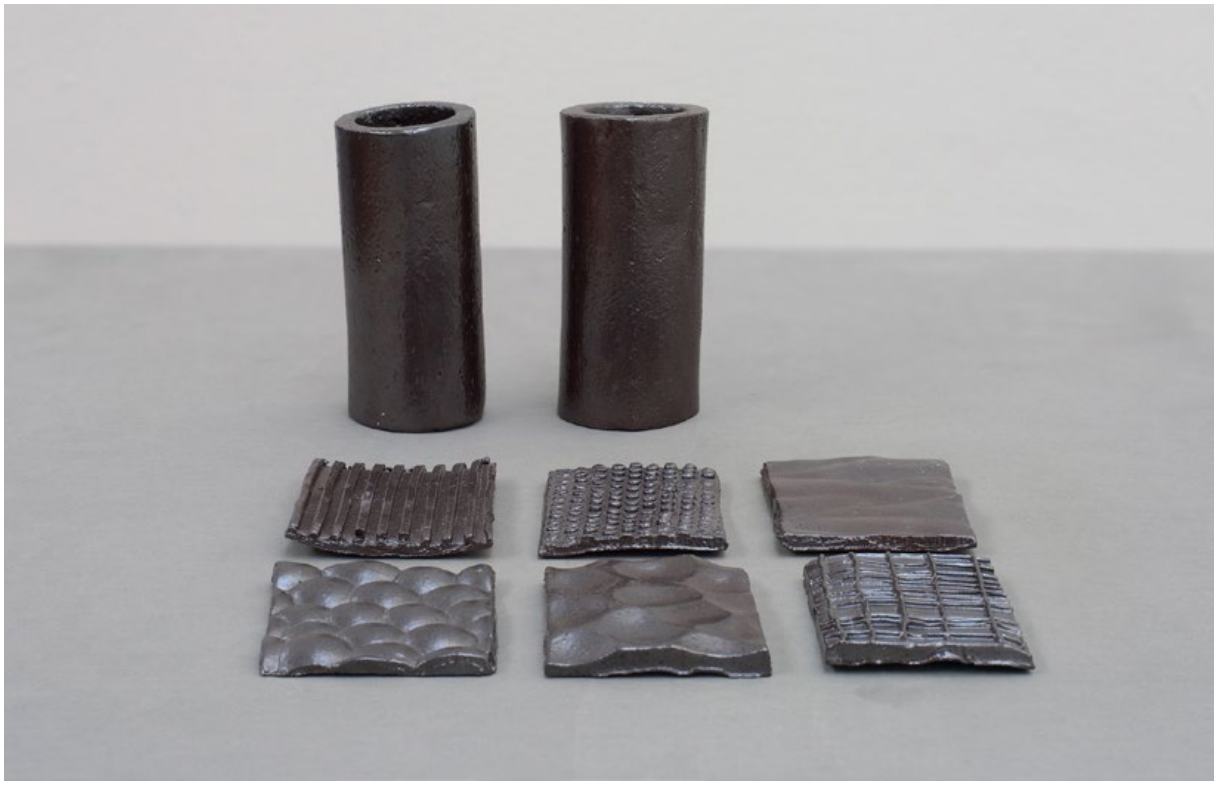
13 Telephone interview, 2024-03-15, Åsa Orrmell, owner of Wallåkra salt glazing factory.



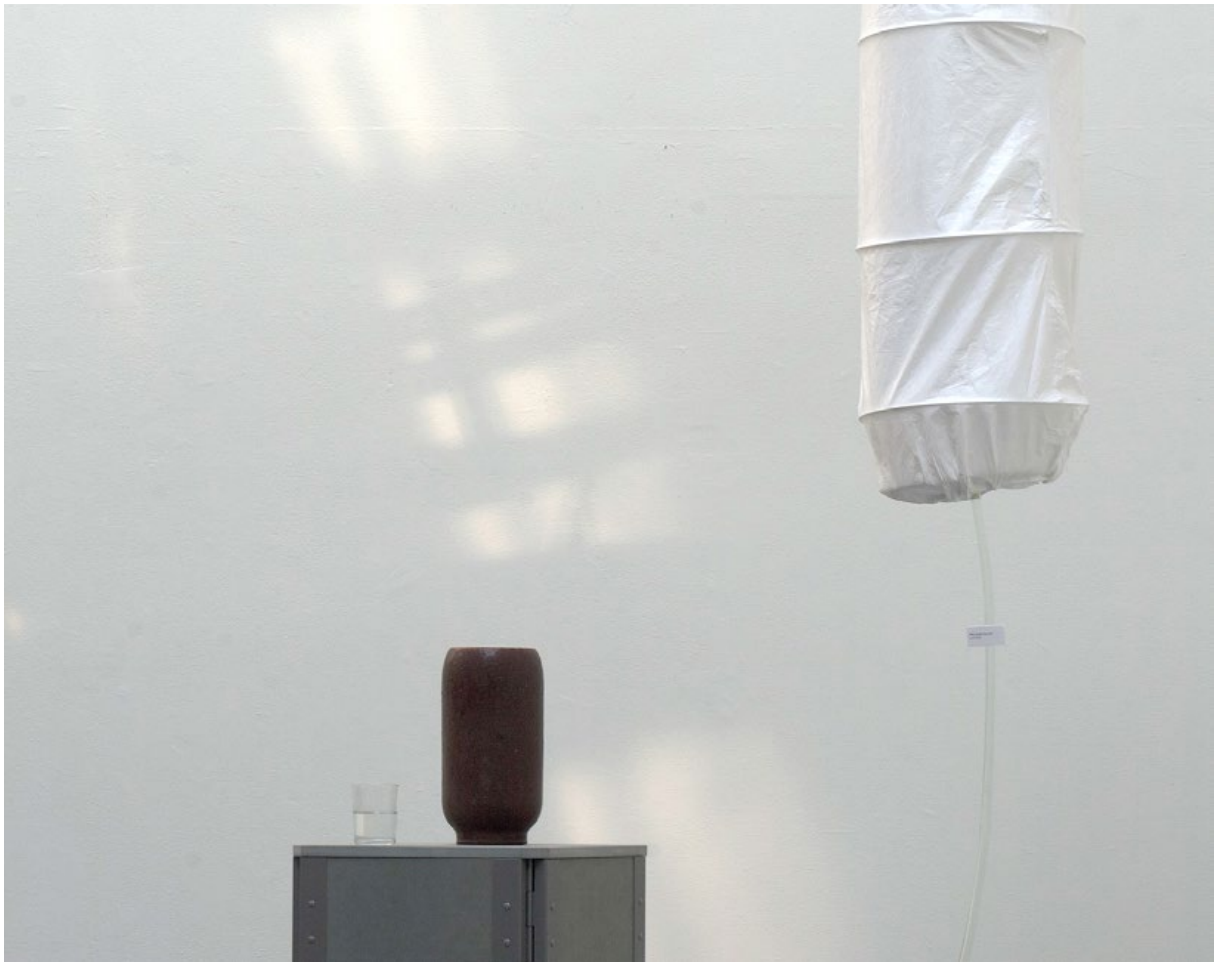
Picture 8. Slipcast molds and stone ware vessels.



Picture 9. LED-light indicating when water is safe to fetch.



Picture 10. Salt glazed samples made in Wallåkra factory (in Skåne).



Picture 11. Rapid prototype of hanging water storage.



Picture 12. Presentation in the exhibition at Konstfack.

Appendix

Learning outcomes from the exhibition
The feedback from visitors to the exhibition was positive when they received a full explanation orally and had the chance to ask questions. When the exhibition was empty, and visitors had to rely on the signs and the artifacts presented, it didn't work as well. There was confusion regarding the actual technique; some misunderstood it as a water kettle made for boiling water. The major surprise was the fact that a fair number of visitors thought the final prototype worked, which was a testament to the craftsmanship. The exhibition was expanded with new artifacts during the week. One water storage unit, which had been discussed during a final presentation, was produced and displayed in the exhibition to test the concept. This added depth and richness to the whole project, even though the design wasn't thoroughly thought out or well executed.

Learning outcomes from the process
The process of producing the final prototypes was very rewarding and, at the same time, challenging. The technique used for the ceramic vessels was slip casting. This technique involves making a positive model, which is then cast in plaster. After that, you have a negative plaster mold in which you pour the liquid stoneware. The plaster then absorbs water from the stoneware, leaving a shell of leather-hard clay, which is then treated as regular clay in the kilns.

General feedback
One interesting piece of feedback was given by Professor Martin Avila, who pointed out that this acoustic technique could be set to different frequencies, allowing control over which pollutants to remove from the water. This was very interesting and was mentioned early in the process but was overlooked due to the strict adherence to NASA technology (which only pushed the water molecules through the nano filter and left everything else behind).

Other interesting feedback and discussions were held with the main tutor, Loove Broms, on what theoretical framework to use in this project. The goal was always to create a realistic and well-thought-out design object that would create a desire to have it, thereby getting the

visitor to accept the underlying concept of a potential future where there is no tap water, or at least not on a regular basis. On that note, different theories were discussed. Was this a classic example of Human-Centered Design (HCD)[14] since it focused so much on the user journey and the different technical aspects of the product? Or was it a more subtle version of speculative[15] and critical design[16] since it proposed an alternative to the probable future where we have tap water and, at most, filter it before it enters our kitchen?

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15 Malpass, M. (2017). ”Critical Design in Context: History, Theory, and Practices.” Bloomsbury Visual Arts, p. 45-63, (2024-06-01)
16 Dunne, A., & Raby, F. , 2013, ”Speculative Everything: Design, Fiction, and Social Dreaming”, MIT Press, p. 34-52, (2024-06-01)

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